MANIPAL UNIVERSITY JAIPUR

Deep Learning (DSE3120) — Course Project Report

AttriSense

Employee Attrition Risk Prediction and Explanation

Predict → Explain → Simulate

Item	Details
Subject	Deep Learning
Course Code	DSE3120
Section	VB
Institution	Manipal University Jaipur

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Prototype developed and tested in a Google Colab notebook

1. Introduction

Employee attrition means employees leaving an organization. When employees leave, the organization may face recruitment costs, training costs, loss of experience and disruption to ongoing work. If the employees who might be at risk of leaving can be identified early, an organization can consider whether some early support is appropriate.

AttriSense is a prototype that uses an Artificial Neural Network (ANN) to estimate employee attrition risk from employee-related information. The project follows three ideas:

- **Predict:** an ANN estimates the probability that an employee belongs to the Leave class.
- **Explain:** SHAP is used to show which input features influenced the model's prediction.
- **Simulate:** What-If analysis tests how the prediction changes when one input value is hypothetically modified.

AttriSense was developed and tested in a notebook. It is a course prototype and not a deployed HR product.

2. Problem Statement and Aim

Problem statement: Organizations may find it difficult to identify employees who could be at risk of leaving. A prediction alone is often not enough, because users also want to understand the factors behind it.

Aim: To build a simple system that predicts attrition risk, explains the model's output, and allows hypothetical changes to employee features to be tested.

3. Objectives

1. Prepare the employee dataset for use in an ANN.
2. Build and train a feed-forward ANN to estimate the probability of an employee leaving.
3. Handle class imbalance using class weights, so that the less common Leave class gets more attention.
4. Evaluate the model using accuracy, precision, recall, F1 score and a confusion matrix, with special focus on Leave-class recall.
5. Convert predicted probabilities into Low, Medium and High risk levels.
6. Explain predictions using SHAP (global and individual explanations).
7. Run What-If simulations using the trained ANN.

4. Dataset

The dataset file is **employee_attrition_hr_2026.xlsx**. The main facts about it are given in Table 1.

Property	Details
Number of records	5,000 employees
Original columns	29
Missing values	None found in the initial check
Target column	attrition (No = 0, Yes = 1)
Class distribution	Stayed (No): 4,105 | Left (Yes): 895
Removed column	employee_id (an identifier, not a meaningful feature)
Input columns used	27 (20 numerical + 7 categorical)

Table 1: Dataset summary

Numerical features include age, monthly income, overtime hours per week, years at company, manager support score, burnout score, engagement score, work-life balance score, performance rating and perceived AI job risk. **Categorical features** are gender, marital status, education level, department, job role, work mode, and whether the employee uses AI tools at work.

5. Methodology

5.1 Data split

The data was split 70:15:15 into training, validation and test sets. The split was stratified, which means each part keeps about the same share of Stay and Leave records as the full dataset.

Dataset	Records	Purpose
Training set	3,500	Used to train the ANN
Validation set	750	Used to monitor the model during training
Test set	750	Used to evaluate the trained model

Table 2: Data split

5.2 Preprocessing

A neural network can only work with numbers, and it learns best when numbers are on similar scales. For example, monthly income is in thousands while a burnout score is a small number; without scaling, large values could dominate learning. Two steps were used:

- **StandardScaler (numerical features):** rescales each numerical column so that it has an average of 0 and a similar spread. This puts all numerical features on comparable scales.
- **OneHotEncoder (categorical features):** turns text categories such as “Engineering” into separate 0/1 columns that the ANN can process.

The preprocessing was fitted on the training set only and then applied to the validation and test sets. This avoids leaking information from the test data into training. After preprocessing, the 27 input columns became **69 processed features**.

5.3 ANN architecture

A feed-forward ANN was built with the Keras Sequential API. Information moves in one direction: from the 69 input features, through two hidden layers, to one output neuron.

Layer	Output size	Parameters	Purpose
Input	69	—	The 69 processed features
Dense + ReLU	64	4,480	First hidden layer; learns patterns
Dropout (0.3)	64	0	Randomly switches off 30% of neurons during training
Dense + ReLU	32	2,080	Second hidden layer
Dropout (0.3)	32	0	Reduces overfitting
Dense + Sigmoid	1	33	Outputs a probability between 0 and 1

Table 3: ANN architecture (total trainable parameters: 6,593)

Components in simple words:

- **ReLU activation:** keeps positive values and turns negative values to zero. It lets the network learn patterns that are not simply straight lines.

- **Dropout (0.3):** during training, 30% of neurons are randomly ignored in each step. This stops the network from memorizing the training data (overfitting).
- **Sigmoid activation:** squeezes the final output into a value between 0 and 1, which we read as the probability of the Leave class.
- **Adam optimizer:** the method that updates the network's weights step by step to reduce the error.
- **Binary cross-entropy loss:** measures how far the predicted probabilities are from the actual 0/1 labels. Training tries to make this smaller.
- **Early stopping:** training stops when validation loss has not improved for 10 epochs, and the best weights seen are restored. Training was allowed up to 100 epochs with a batch size of 32.

5.4 Class weighting and risk levels

Only 895 of the 5,000 records are Leave cases. A model trained on such data can pay most attention to the majority Stay class. To reduce this effect, **class weights** were used (computed with the "balanced" option): Stay = 0.609 and Leave = 2.791. A mistake on a Leave employee therefore counts about 4.6 times more than a mistake on a Stay employee during training.

Why recall is the priority. AttriSense is an early-warning project. A **false negative** is an employee who actually leaves but is predicted to Stay; in an early-warning system this means an at-risk employee may be missed. A higher Leave-class **recall** means the model finds a larger share of the employees who really leave, so recall for the Leave class is the key measure for this goal. Accuracy is still useful, but it is not the main measure here: because most employees stay, a model can show fairly high accuracy while missing many leavers. Class weighting increases the model's attention to the Leave class, although it can also create more false alarms. For a balanced picture, precision, F1 score, accuracy and the confusion matrix are all reported.

Risk levels. The predicted probability was converted into three levels. These thresholds were chosen for this project and are not universal industry standards.

Risk level	Predicted probability
Low	Below 30%
Medium	30% to below 60%
High	60% or above

Table 4: Risk level thresholds

5.5 SHAP

SHAP (SHapley Additive exPlanations) explains a model's output by giving each input feature a score showing how much it pushed the prediction up or down. We used SHAP's PermutationExplainer on the weighted ANN with the test set. Two views were produced: a **global summary plot** (which features matter most across many employees) and an **individual waterfall plot** (why one employee received their prediction).

5.6 What-If analysis

What-If analysis uses the already-trained weighted ANN. The steps were:

1. Select an employee's original record.
2. Make a copy of the record.
3. Change one input feature.
4. Apply the same fitted preprocessing steps.
5. Pass the modified data through the same trained ANN.
6. Compare the original predicted probability with the new one.

6. Results and Evaluation

6.1 Training behaviour

The weighted ANN trained for 10 epochs before early stopping ended training. The lowest validation loss (0.5135) occurred at epoch 3, and those best weights were restored for evaluation. Figures 1 and 2 show training and validation loss and accuracy. Training accuracy is lower than validation accuracy partly because dropout is active during training and because the class weights make the training loss stricter on Leave cases.

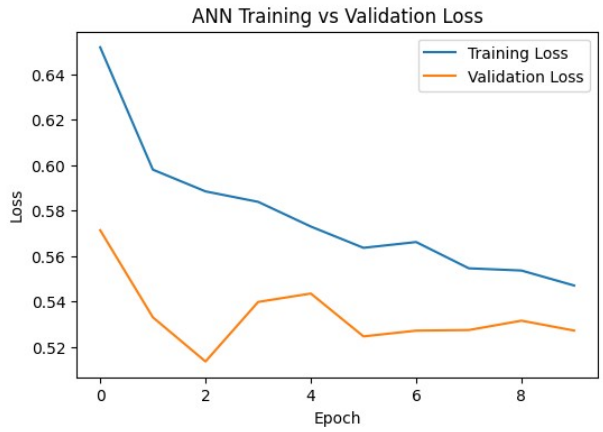


Figure 1: ANN training vs validation loss (weighted ANN)

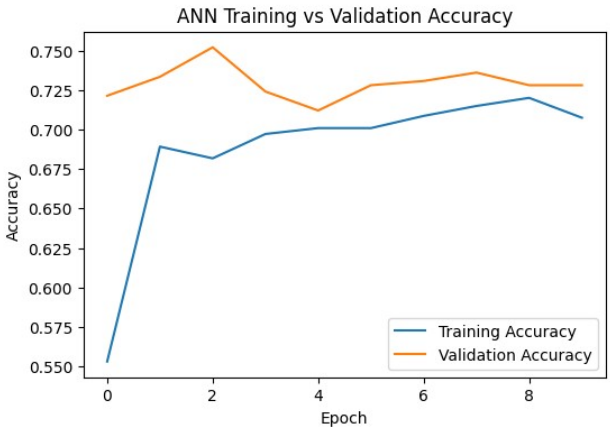


Figure 2: ANN training vs validation accuracy (weighted ANN)

6.2 Metrics on the test set

All metrics below were measured on the 750-record test set, using a threshold of 0.5 on the weighted ANN's output probability.

Metric	Result	Meaning in simple words
Accuracy	70.67%	Out of all 750 employees, the share predicted correctly (530 of 750).
Precision (Leave)	33.96%	When the model says “Leave”, it is right about 1 time in 3.
Recall (Leave)	67.91%	Of the 134 employees who actually left, the model found 91.
F1 score (Leave)	45.27%	A single score that balances precision and recall.

Table 5: Weighted ANN test results

Recall is the metric most closely tied to the project goal. The model correctly flagged **91 of the 134 actual leavers**, a recall of 67.91%. Precision is lower, which means that some employees flagged as Leave actually stay. This is the usual trade-off when a model is tuned to catch more leavers.

6.3 Confusion matrix

Figure 3 shows the confusion matrix of the weighted ANN. Table 6 gives the same counts with their meaning.

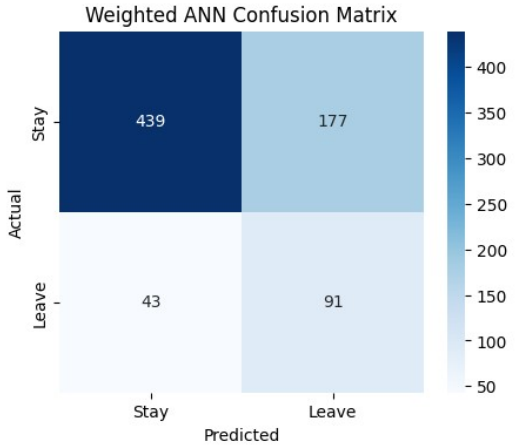


Figure 3: Weighted ANN confusion matrix (test set, 750 records)

Actual class	Predicted Stay	Predicted Leave
Stay	439 (correct)	177 (false alarms)
Leave	43 (missed leavers)	91 (correct)

Table 6: Confusion matrix counts

- **439** employees actually stayed and were correctly predicted as Stay.
- **177** employees actually stayed but were predicted as Leave (false positives, or false alarms).
- **43** employees actually left but were predicted as Stay (false negatives, or missed leavers).
- **91** employees actually left and were correctly predicted as Leave.

Effect of class weighting. The same ANN trained without class weights reached a higher accuracy of 83.47%, but its Leave-class recall was only about 16% (21 of 134 leavers found; Figure 4). Class weighting raised recall from 21 to 91 correct leavers, at the cost of more false alarms and lower overall accuracy. This is why accuracy alone would not describe the project goal well.

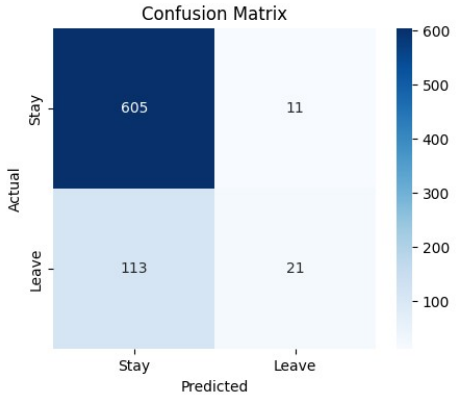


Figure 4: Confusion matrix of the same ANN trained without class weights (for comparison)

6.4 Risk distribution

The predicted probabilities for the 750 test employees were converted into risk levels using the thresholds in Table 4 (Figure 5).

Risk level	Number of employees
Low	187
Medium	416
High	147
Total	750

Table 7: Risk level distribution on the test set



Figure 5: Employee attrition risk distribution (test set)

These risk levels are model predictions, not confirmed future outcomes.

6.5 SHAP explanations

Global explanation. Figure 6 is the SHAP summary plot. Each dot is one employee. Features at the top influenced the model's output the most. Dots to the right pushed the predicted risk higher; dots to the left pushed it lower. The colour shows whether the feature value was high (red) or low (blue). The features near the top were burnout score, manager support score, engagement score, last review score and overtime hours per week. Performance rating, number of prior companies, work-life balance score, business travel days, perceived AI job risk, training hours and salary hike percentage also appear among the leading features.

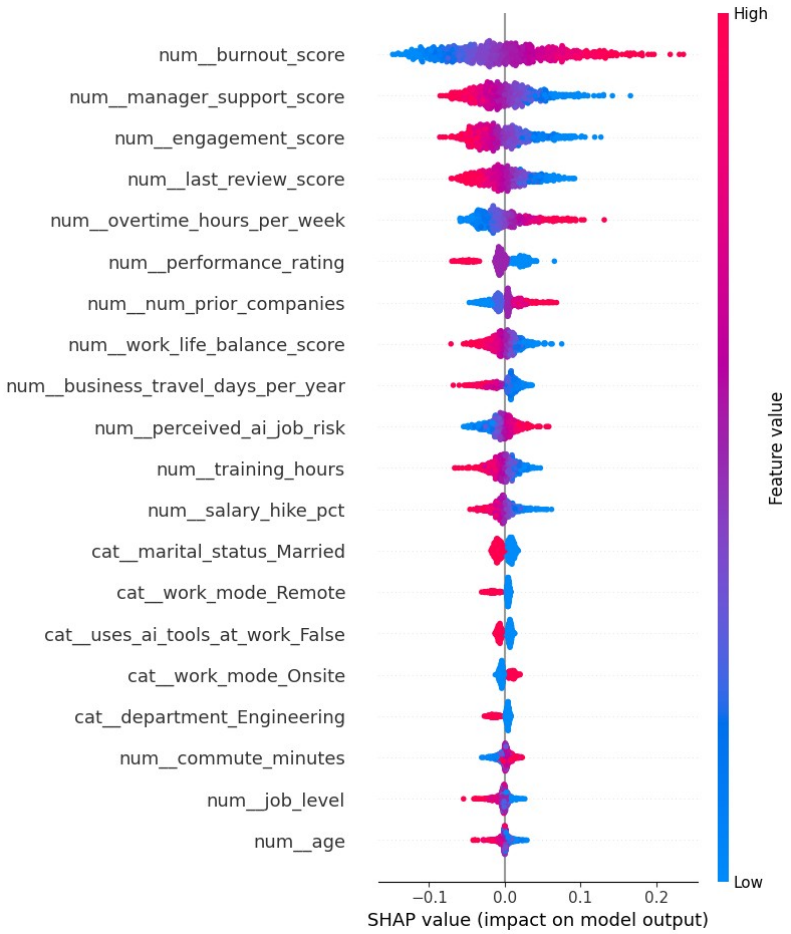


Figure 6: Global SHAP summary plot (weighted ANN, test set)

Individual explanation. Figure 7 is a SHAP waterfall plot for one employee (test record 0: a 40-year-old Data Engineer in the Engineering department). The plot starts from the model's average output, E[f(X)] = 0.441, and each bar shows how one feature moves the prediction up (red) or down (blue) until it reaches this employee's predicted probability, f(x) = 0.209 (Low Risk). For this employee, a high engagement score, a low burnout score, a high last review score and a high manager support score each pushed the risk down. Overtime hours and marital status pushed it up slightly.

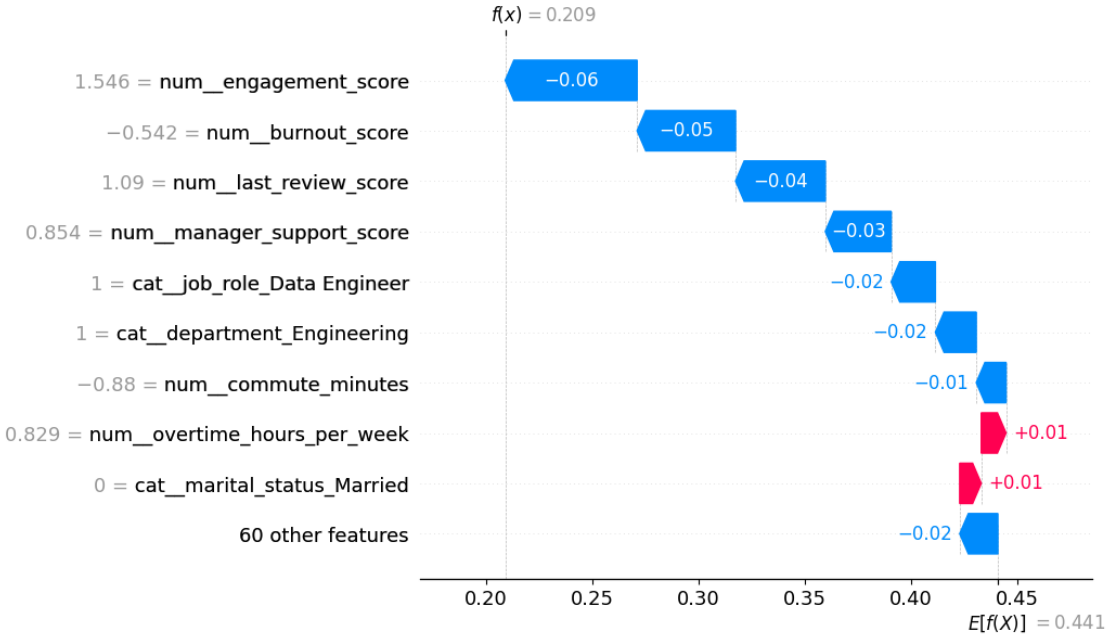


Figure 7: Individual SHAP waterfall plot (test record 0)

SHAP makes the model easier to interpret, but it explains how the **model** behaves. It does not prove that any feature causes an employee to leave.

6.6 What-If results

The What-If analysis was run for the same employee as in Figure 7 (original predicted risk 20.91%).

Hypothetical change	Original risk	New risk	Difference
Overtime reduced from 10.5 to 5 hours per week	20.91%	21.10%	+0.19 percentage points
Engagement score increased from 8.8 to 9.5	20.91%	19.78%	−1.13 percentage points

Table 8: What-If simulation results

In the first example, reducing overtime changed the model's output by a very small amount (about 0.2 percentage points), and in the upward direction. This shows that a single feature does not always move the prediction the way one might expect, because the ANN combines many features together. In the second example, increasing engagement lowered the predicted risk by about 1.1 percentage points.

These are **hypothetical changes in model output**. They do not prove that reducing overtime or raising engagement will change real attrition, and no actual HR intervention was carried out.

7. Work Completed

The following components have been implemented in the project notebook:

Area	Completed work
Data	Dataset loading and inspection; initial missing-value check; target encoding; train/validation/test split
Preprocessing	Numerical standardization (StandardScaler); categorical one-hot encoding (OneHotEncoder)
Model	ANN construction, training and evaluation; class weighting; prediction probabilities
Risk	Low, Medium and High risk labels
Explainability	Global SHAP summary explanation; individual SHAP waterfall explanation
Simulation	What-If simulations for overtime and engagement
Visuals and results	Training/validation loss and accuracy graphs; confusion matrix; risk distribution chart; final results table

Table 9: Summary of completed work

8. Current Status and Optional Future Improvements

The core AttriSense prototype is implemented: it predicts attrition risk, explains predictions with SHAP, and supports What-If simulation. No further application, deployment or additional model is required for the prototype as described. If time permits, the following would be useful optional improvements:

- Running the notebook from top to bottom once more and checking that all reported numbers come from the same final run.
- Testing the model on additional or newer data.
- Reviewing the risk thresholds (30% and 60%).

- Reviewing privacy, fairness and possible bias before any real-world HR use.

9. Conclusion

AttriSense combines three parts in one prototype: ANN-based attrition prediction, SHAP-based explanation, and What-If simulation. The ANN (69 → 64 → 32 → 1) was trained with class weights to give more attention to employees who leave.

Because AttriSense is an early-warning project, Leave-class recall was prioritized. On the test set, the weighted ANN found 91 of 134 actual leavers (recall 67.91%), with accuracy of 70.67%, precision of 33.96% and F1 score of 45.27%. The trade-off is that 177 employees who stayed were flagged as Leave. The unweighted ANN had higher accuracy (83.47%) but found only 21 of the 134 leavers, which shows why recall was the better guide for this goal.

The main contribution of the project is the practical combination of **Predict → Explain → Simulate** in one working notebook prototype. We did not create a new ANN algorithm, and the system is not a production-ready HR tool. SHAP and What-If results describe the behaviour of the model; they do not prove cause and effect.

10. Short Summary

Item	Summary
Project	AttriSense: employee attrition risk prediction and explanation
Data	5,000 records, 27 input features (69 after preprocessing); 4,105 Stay and 895 Leave
Model	Feed-forward ANN 69 → 64 → 32 → 1, ReLU, Dropout 0.3, Sigmoid, Adam, binary cross-entropy, early stopping, class weights
Test results	Leave recall 67.91%; accuracy 70.67%; precision 33.96%; F1 45.27%
Confusion matrix	439 TN, 177 FP, 43 FN, 91 TP
Risk levels	Low 187, Medium 416, High 147 (750 test employees)
Explain	SHAP summary and waterfall plots
Simulate	Overtime 10.5 → 5: +0.19 points; engagement 8.8 → 9.5: −1.13 points (hypothetical)
Status	Core prototype implemented; optional future improvements listed in Section 8